SOUMYA BANERJEE

CONTACT INFORMATION

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EDUCATION

PhD in Computer Science, 2013, University of New Mexico, USA
Bachelor of Engineering (Computer Science) with Distinction, 2003, Nagpur University, India

RECENT EMPLOYMENT

Assistant Research Professor, University of Cambridge, UK, 2025 till date
Project description: I develop explainable AI techniques and apply them to various domains like healthcare. I also teach courses on AI and supervise student projects.

Senior Research Fellow and Affiliated Lecturer, University of Cambridge, UK, 2022 till date
Project description: I develop explainable AI techniques and apply them to various domains like healthcare. I also teach courses on AI and supervise student projects.

Postdoctoral Researcher, University of Cambridge, UK, 2019 to 2022
Project description: I used machine learning techniques and data science on electronic healthcare record data to help stratify patients and predict mortality. This has been published in *Nature Partner Journal Schizophrenia*.

Postdoctoral Researcher, University of Oxford, UK, 2016 to 2018
Project description: The project involves modeling negative selection in the immune system. I use multi-scale models to simulate how immune system cells are trained in the thymus. The work resulted in a manuscript published in the *Journal of the Royal Society Interface*.

Researcher, Commonwealth Scientific and Industrial Research Organisation, Australia, 2015 to 2016
Project description: The project involved modeling socio-economic systems by combining open data with machine learning techniques. We used a publicly available dataset from the UCI Machine Learning Repository on crime in cities. Using statistical and machine learning models combined with non-linear dynamical models we inferred novel patterns in crime in cities. This resulted in a manuscript published in the journal *Palgrave Communications*.

Post-doctoral research fellow, Harvard University, Harvard Medical School and Broad Institute of Harvard and MIT, USA, 2014 to 2015
Project description: The work involved investigating the interaction between the immune system and the human microbiome. The human microbiome has been implicated in various disorders like inflammatory bowel disease. I used computational techniques to predict compounds that are likely to be secreted by microbes in the human gut based on their genome. Compounds that are computationally predicted to be differentially abundant in healthy vs. disease subjects across multiple cohorts are likely to play an important role in inflammatory bowel disease. This project was implemented on python, shell scripts and MATLAB.

Post-doctoral research fellow, Max Planck Institute for Molecular Physiology, Germany, 2013 to 2014
Project description: The Rho GTPase family of proteins influence cell motility and cytoskeleton dynamics across all eukaryotes. I investigated the interaction between three different Rho GTPase

proteins and how this shapes cell motility. I used non-linear differential equation models to simulate the interaction between the different Rho GTPase proteins. In order to make more accurate estimates of model parameters despite noisy experimental data, I used hierarchical Bayesian models to combine data from different experiments. All code was implemented in MATLAB and shell scripts and run on high-performance computing platforms. I also implemented automated image analysis of cell images from microscopes using tools like CellProfiler and ImageJ.
I was also trained in various wet-lab skills like plating and splitting cells, staining with primary and secondary antibodies, transfection and microscopy.
The work resulted in a publication in the journal *Angewandte Chemie* (Impact Factor = 13.1) and the *Journal of Cell Biology* (Impact Factor = 9.7). One more manuscript is in preparation.

Research Intern, Los Alamos National Laboratories, Theoretical Biology and Biophysics Group, USA, 2009 to 2012
Project description: The project (PhD thesis) involved modeling emerging pathogens (like West Nile virus and avian influenza) that have sparse experimental data and considerable uncertainty in parameters that go into models. I simulated the complex interaction between the virus, host cells and the host immune system using non-linear differential equations. I built mathematical models that are able to predict biologically relevant quantities like the basic reproductive number and infectious virion burst size with a lot of precision, despite noisy data and uncertainty in inputs to our models. Our novelty lay in:
1) A combination of knockout experimental data and mathematical models, and
2) Hierarchical Bayesian models that incorporate hierarchy in data and biologically motivated priors to generate more precise estimates than otherwise possible from sparse experimental data.
The work resulted in a manuscript accepted in the *Journal of the Royal Society Interface*.
I also presented the work in three international venues. The MATLAB programs to solve these complex statistical problems was made accessible using easy to use graphical user interfaces and made openly available. Based on the number of times these tools have been downloaded, I have been ranked within the Top 250 worldwide as a MATLAB open-source code contributor.

Research Assistant, University of New Mexico, Department of Computer Science, USA, 2007 to 2013
Project description: The project involved work in computational immunology: using techniques from computer science to solve problems in immunology and taking inspiration from the immune system to solve problems in computer science. Specifically, my work suggested how the physical architecture of the immune system might lead to nearly scale-invariant immune search and response. Using the immune system as an inspiration I proposed architectures for human-engineered distributed systems (like peer-to-peer networks and mobile ad-hoc networks) with faster search and response characteristics.
The work resulted in 1 journal publication (*Swarm Intelligence*) and 4 peer-reviewed conference publications.

Senior Software Engineer and Technical Lead, Cognizant Technology Solutions Ltd., India, 2004 to 2007
Project description: I worked on projects in the financial services sector with various Fortune 500 clients. The work involved interacting with the client, defining project requirements, implementing solutions according to requirements, coordinating a team of developers, and rigorous testing. I worked on the UNIX platform and used C, C++, shell scripts and Sybase to implement solutions. I also worked on developing web-based applications with ASP.NET and database development with MS SQL Server.

Software Engineer, Automation Engineers, India, 2003 to 2004

TEACHING EXPERIENCE

I am a Fellow of the Higher Education Academy, Advance HE. I teach machine learning at the University of Cambridge Department of Computer Science and Bioinformatics Training Centre. I have also taught students from Uganda through the University of Cambridge Africa programme. I taught data science at the University of Oxford Complex Networks Summer School.

My experience in teaching involves teaching data science, machine learning, advanced machine learning topics, reproducibility techniques in machine learning and data visualization. I have been teaching in higher education institutions for the last 10 years.

An example of a course that I have developed and taught is here:

https://cambiotraining.github.io/ml-unsupervised/

I have taught Unconventional approaches to Artificial Intelligence. This is a special topics course I developed. The teaching material is available online here:

https://github.com/neelsoumya/special_topics_unconventional_AI

I have also developed materials for and taught a course on reproducible research in R:

https://github.com/neelsoumya/teaching_reproducible_science_R

I also teach data science courses and visualization. Some course materials are available here:

https://github.com/neelsoumya/visualization_lecture

I have taught machine learning, complex systems, and basics of programming. All my teaching material is available here:

https://sites.google.com/site/neelsoumya/teaching

Other teaching activities are outlined below:
1) Cambridge-Africa Programme, University of Cambridge, 2020 (lectures)
2) Oxford Summer School in Economic Networks, Mathematical Institute, University of Oxford, 2017 (lectures and tutorials)
3) Trained in teaching at the University of Oxford Doctoral Training Center, 2017
4) Lectured at the Complex Adaptive Systems course in Department of Computer Science, University of New Mexico, USA, 2009-2012 (lectures and tutorials on complex systems and computer science)
5) Co-designed teaching resources with educators to make my teaching broadly accessible (samples of my teaching material are available online https://www.simiode.org/resources/3206 and https://osf.io/25gnz/)

PUBLICATIONS

NOTE: In my field (Computer Science), people submit to conferences and conferences are peer-reviewed

Key Publications

1) Patient and public involvement to build trust in artificial intelligence: a framework, tools and case studies, S. Banerjee, P. Alsop, L. Jones, R. Cardinal, *Patterns*, 3(6):100506, 2022
(*Cell Press publishing group*)
2) A class-contrastive human-interpretable machine learning approach to predict mortality in severe mental illness, S. Banerjee, P. Lio, P. Jones, R. Cardinal, *Nature Partner Journal Schizophrenia*, 7, 60, 2021
(*Nature publishing group, Impact Factor = 6.3*)
3) Deconvolution of monocyte responses in inflammatory bowel disease reveals an IL-1 cytokine network that regulates IL-23 in genetic and acquired IL-10 resistance, D. Aschenbrenner, M. Quaranta, S. Banerjee, et al., *Gut,* 2020
(*British Medical Journal publishing group, Impact Factor = 19.8*)
4) Hydroxychloroquine: balancing the needs of LMICs during the COVID-19 pandemic, S. Banerjee, *Lancet Rheumatology*, 2(7):385-386, 2020
(*Lancet publishing group*)
5) Influence of correlated antigen presentation on T cell negative selection in the thymus, S. Banerjee, S.J. Chapman, *Journal of the Royal Society Interface*, 15(148), 20180311, 2018
(*Royal Society publishing group, Impact Factor = 4.3*)

Refereed Journal Papers

1) Balancing Objects and Processes: Advocating Pluralism in Biology: Comment on "Thoughts and thinkers: On the Complementarity between Objects and Processes" by Chris Fields and Michael Levin, S. Banerjee, *Physics of Life Reviews*, 2025
(Impact Factor 13.7)
2) DataSHIELD: Mitigating disclosure risk in a multi-site federated analysis platform, D. Avraam, R. Wilson, N. Chan, S. Banerjee et al. *Bioinformatics Advances*, vbaf046, 2025
(Oxford University Press publishing group)
3) Neural networks for abstraction and reasoning, M. Bober-Irizar, S. Banerjee, *Scientific Reports*, 14, 27823, 2024
(Nature Publishing Group, Impact Factor 3.8)
4) Generating Complex Explanations for Artificial Intelligence Models: An Application to Clinical Data on Severe Mental Illness, S. Banerjee, *Life*, 14(7), 807, 2024
5) Intelligence in animals, humans and machines: a heliocentric view of intelligence? H. Holm, S. Banerjee, *AI & Society*, 2024
6) Involving patients in artificial intelligence research to build trustworthy systems, S. Banerjee, S. Griffiths, *AI & Society*, 2023
7) dsSurvival 2.0: Privacy enhancing survival curves for survival models in the federated DataSHIELD analysis system, S. Banerjee, T. Bishop, *BMC Research Notes* 16, 98, 2023
8) Patient and public involvement to build trust in artificial intelligence: a framework, tools and case studies, S. Banerjee, P. Alsop, L. Jones, R. Cardinal, *Patterns*, 3(6):100506, 2022
(*Cell Press publishing group*)
9) Software Application Profile: ShinyDataSHIELD—an R Shiny application to perform federated non-disclosive data analysis in multicohort studies, X. Escribà-Montagut, Y. Marcon, D. Avraam, S. Banerjee, T. Bishop, P. Burton, J. González, *International Journal of Epidemiology,* dyac201, 2022
(*Oxford University Publishing Group, Impact Factor 9.8*)

10) dsSynthetic: Synthetic data generation for the DataSHIELD federated analysis system, S. Banerjee, T. Bishop, *BMC Research Notes*, 15(1):230, 2022
11) dsSurvival: Privacy preserving survival models for federated meta-analysis of clinical data, S. Banerjee, et al, *BMC Research Notes*, 15(1):197, 2022
12) A class-contrastive human-interpretable machine learning approach to predict mortality in severe mental illness, S. Banerjee, P. Lio, P. Jones, R. Cardinal, *Nature Partner Journal Schizophrenia*, 7, 60, 2021
(*Nature publishing group, Impact Factor = 6.3*)
13) Optogenetic tuning reveals Rho amplification-dependent dynamics of a cell contraction signal network, D. Kamps, J. Koch, V. Juma, E. Campillo, M. Graessl, S. Banerjee, T. Mazel, X. Chen, Y. Wu, S. Portet, A. Madzvamuse, P. Nalbant, L. Dehmelt, *Cell Reports*, 33(9):108467, 2020
(*Cell Press publishing group, Impact Factor = 8.1*)
14) Deconvolution of monocyte responses in inflammatory bowel disease reveals an IL-1 cytokine network that regulates IL-23 in genetic and acquired IL-10 resistance, D. Aschenbrenner, M. Quaranta, S. Banerjee, et al., *Gut,* 2020
(*British Medical Journal publishing group, Impact Factor = 19.8*)
15) Hydroxychloroquine: balancing the needs of LMICs during the COVID-19 pandemic, S. Banerjee, *Lancet Rheumatology*, 2(7):385-386, 2020
(*Lancet publishing group*)
16) The early impact of COVID-19 on mental health and community physical health services and their patients' mortality in Cambridgeshire and Peterborough, UK, S. Chen, P. Jones, B. Underwood, A. Moore, E. Bullmore, S. Banerjee, et al., *Journal of Psychiatric Research*, 131, 244-254, 2020
(*Impact factor = 4.4*)
17) Predictive metabolomic profiling of microbial communities using amplicon or metagenomic sequences, H. Mallick, E. Franzosa, L. McIver, S. Banerjee, A. Sirota-Madi, A. Kostic, C. Clish, H. Vlamakis, R. Xavier, C. Huttenhower, *Nature Communications*, 10(1):3136, 2019
(*Nature publishing group, Impact Factor = 12.1*)
18) Influence of correlated antigen presentation on T cell negative selection in the thymus, S. Banerjee, S.J. Chapman, *Journal of the Royal Society Interface*, 15(148), 20180311, 2018
(*Royal Society publishing group, Impact Factor = 4.3*)
19) Modelling the effects of phylogeny and body size on within-host pathogen replication and immune response, S. Banerjee, A. Perelson, M. Moses, *Journal of the Royal Society Interface*, 14(136), 20170479, 2017
(*Royal Society publishing group, Impact Factor = 4.3*)
20) An excitable Rho GTPase signaling network generates dynamic subcellular contraction patterns, M. Graessl, J. Koch, A. Calderon, S. Banerjee, T. Mazel, N. Schulze, J. Jungkurth, A. Koseska, L. Dehmelt, P. Nalbant, *Journal of Cell Biology*, 216(12), 4271-4285, 2017
(*Cell publishing group, Impact Factor = 9.7*)
21) Towards a Quantitative Understanding of Within Host Dynamics of West Nile Virus Infection, S. Banerjee, J. Guedj, R. Ribeiro, M. Moses & A. Perelson, *Journal of the Royal Society Interface*, 13(117), 20160130, 2016
(*Royal Society publishing group, Impact Factor = 4.3*)
22) A spatial model of the efficiency of T cell search in the influenza infected lung, D. Levin, S. Forrest, S. Banerjee, M. Moses & F. Koster, *Journal of Theoretical Biology*, 398(7), 52-63, 2016
(*Impact Factor = 2.2*)
23) Competitive dynamics between criminals and law enforcement explains the super-linear scaling of crime in societies, S. Banerjee, Hentenryck, P.V. & Cebrian, M., *Palgrave Communications*, 1(1), 15022, 2015
(*Nature publishing group*)

24) A bioorthogonal small-molecule switch system for controlling protein function in cells, P. Liu, A. Calderon, G. Konstantinidis, J. Hou, S. Voss, X. Chen, F. Li, S. Banerjee, J. Hoffmann, C. Theiss, L. Dehmelt & Y. Wu, *Angewandte Chemie*, 53(38), 10049-10055, 2014
(*Impact Factor = 13.7*)
25) Science and technology consortia in US biomedical research: A paradigm shift in response to unsustainable academic growth, C. Balch, H. Arias-Pulido, S. Banerjee, A. Lancaster, K. Clark, M. Perilstein, B. Hawkins, J. Rhodes, P. Sliz, J. Wilkins and T. Chittenden, *BioEssays*, 37(2), 119-122, 2014
(*Impact Factor = 5.4*)
26) Scale Invariance of Immune System Response Rates and Times: Perspectives on Immune System Architecture and Implications for Artificial Immune Systems, S. Banerjee & M. Moses, *Swarm Intelligence*, 4(4), 301-308, 2010
(*Impact Factor = 2.1*)

Refereed Workshop Papers
1) Bridging the Gap with Retrieval-Augmented Generation: Making Prosthetic Device User Manuals Available in Marginalised Languages Ogbonna Ikechukwu, Lesley Davidson, Soumya Banerjee, Abhishek Dasgupta, Laurence Kenney, Vikranth Harthikote Nagaraja *Data Science Africa 2025 Workshop*, 2025
2) Prime Directives for Responsible AI for Africa: A Manifesto for Inclusive Technology Soumya Banerjee, Vikranth Harthikote Nagaraja, Innocent Nyalala, Nirav Pravinbhai Bhatt *Data Science Africa 2025 Workshop*, 2025
3) Understanding Intelligence Across Humans, Machines, and Animals S. Banerjee, *Workshop on Exploring Interdisciplinary Frontiers: Cognitive Science, Computational Modeling, and Artificial Intelligence*, 2025
4) Towards a theory of life-like systems S. Banerjee, *IEEE Symposium on CI in Artificial Life and Cooperative Intelligent Systems (Late Breaking Abstract): Trondheim, Norway*, 2025
5) Augmenting human intelligence on abstraction and reasoning tasks: a path synergistic AI-human interaction for the Abstraction and Reasoning Corpus, S. Banerjee, *IEEE Symposium on CI in Artificial Life and Cooperative Intelligent Systems (Late Breaking Abstract): Trondheim, Norway*, 2025
6) "I apologize for my actions": Emergent Properties of Generative Agents and Implications for a Theory of Mind, N. Diamond, S. Banerjee *AAAI workshop on Advancing Artificial Intelligence through Theory of Mind (ToM4AI)*, 2025
7) The BACON system for equation discovery from scientific data: Reconciling classical artificial intelligence with modern machine learning approaches, Jonah Miller, S. Banerjee *4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*, 2025
8) Enhancing AI Capabilities on the Abstraction and Reasoning Corpus: A Path Toward Broad Generalization in Intelligence, S. Banerjee, *4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*, 2025
9) AugARC: Augmented Abstraction and Reasoning Benchmark for Large Language Models, K. Bikov, M. Bober-Irizar, S. Banerjee, *AAAI Workshop on Preparing Good Data for Generative AI: Challenges and Approaches*, 2025
10) AI and the Future of the Technosphere: A Path Towards Co-Existence with Superintelligence, S. Banerjee, *AAAI Workshop on Post-Singularity Symbiosis*, 2025
11) MASR: Multi-Agent System with Reflection for the Abstraction and Reasoning Corpus, K. Bikov, M. Bober-Irizar, S. Banerjee, *AAAI 2025 Workshop on Advancing LLM-Based Multi-Agent Collaboration*, 2025
12) What can machines teach us in our journey of reproducing human scientific creativity? A. Mazumdar, N. Karim, S. Banerjee, *IEEE Symposium on CI in Artificial Life and Cooperative Intelligent Systems (Late Breaking Abstract): Trondheim, Norway*, 2025

13) Reflection System for the Abstraction and Reasoning Corpus, K. Bikov, M. Bober-Irizar, S. Banerjee, *AAAI 2nd AI4Research Workshop: Towards a Knowledge-grounded Scientific Research Lifecycle*, 2025
14) On the Ethical Considerations of Generative Agents, N. Diamond, S. Banerjee, *NeurIPS Workshop, Socially Responsible Language Modelling Research (SoLaR)*, 2024
15) Neural networks for abstraction and reasoning, M. Bober-Irizar, S. Banerjee, *NeurIPS Workshop, Multimodal Algorithmic Reasoning (MAR)*, 2024
(selected for oral presentation)
16) Enhancing patient stratification and interpretability through class-contrastive and feature attribution techniques, S. Olowu, N. Lawrence, S. Banerjee, *NeurIPS Workshop, Interpretable AI: Past, Present and Future,* 2024
17) Building Trustworthy AI: The Role of Patient and Public Involvement in Healthcare AI Development, S. Banerjee *Proceedings of the AAAI Symposium Series* 4 (1), 329-331, 2024
18) The BACON system for equation discovery from scientific data: Transforming classical artificial intelligence with modern machine learning approaches, J. Miller, S. Banerjee, *Association for the Advancement of Artificial Intelligence (AAAI) Fall Symposium Series, Integrated Approaches to Computational Scientific Discovery*, 2024

Refereed Conference Papers
1) Evaluating Machine Learning Models for Prediction of Attention Deficit Hyperactivity Disorder among Autistic Individuals Using Genetic Data, *2025 World Congress of Psychiatric Genetics*, 2025
2) Cosmicism and Artificial Intelligence: Beyond Human-Centric AI S. Banerjee *The 1st International Online Conference of the Journal Philosophies*, 2025
3) When Planes Fly Better Than Birds: Should AIs Think Like Humans? S. Banerjee *The 1st International Online Conference of the Journal Philosophies*, 2025
4) Enhancing patient stratification and interpretability through class-contrastive and feature attribution techniques, S. Olowu, N. Lawrence, S. Banerjee, *Proceedings of the 2025 IEEE Symposium Series on Computational Intelligence, Trondheim, Norway*, 2025
5) "I apologize for my actions": Emergent Properties of Generative Agents and Implications for a Theory of Mind, N. Diamond, S. Banerjee *Proceedings of the 2025 IEEE Symposium Series on Computational Intelligence, Trondheim, Norway*, 2025
6) Analysis of a Planetary Scale Scientific Collaboration Dataset Reveals Novel Patterns, S. Banerjee, *Proceedings of the Complex Systems Digital Campus 2015 – World e-Conference, Conference on Complex Systems,* 2016 (peer-reviewed conference)
7) A Multi-Agent System Approach to Load-Balancing and Resource Allocation for Distributed Computing, S. Banerjee and J. Hecker, *Proceedings of the Complex Systems Digital Campus 2015 – World e-Conference, Conference on Complex Systems*, 2016 (peer-reviewed conference)
8) Computationally Simulating Intermodal Terminal Attractiveness and Demand, S. Banerjee et al., *Proceedings of the 23rd World Congress on Intelligent Transport Systems*, 2016
9) Analysis of Demand and Operations of Intermodal Terminals, R. Garcia-Flores, S. Banerjee et al., *Proceedings of the 24th National Conference of the Australian Society for Operations Research*, 2016
10) Forecasting in the era of Big Data: Lessons and Pitfalls, Y. Tyshetskiy, S. Banerjee et al., *Proceedings of the Annual Conference of the International Association of Maritime Economists*, 2016
11) The Value of Inflammatory Signals in Adaptive Immune Responses, S. Banerjee, D. Levin, M. Moses, F. Koster & S. Forrest, *The 10th International Conference on Artificial Immune Systems (ICARIS)*, Lecture Notes in Computer Science, Volume 6825/2011, 1-14, 2011
12) Biologically Inspired Design Principles for Scalable, Robust, Adaptive, Decentralized Search and Automated Response (RADAR), M. Moses & S. Banerjee, *Proceedings of the 2011 IEEE*

Conference on Artificial Life, 30-37, 2011
13) Modular RADAR: An Immune System Inspired Search and Response Strategy for Distributed Systems, S. Banerjee & M. Moses, *The 9th International Conference on Artificial Immune Systems (ICARIS)*, Lecture Notes in Computer Science, Volume 6209/2010, 116-129, 2010
14) A Hybrid Agent Based and Differential Equation Model of Body Size Effects on Pathogen Replication and Immune System Response, S. Banerjee & M. Moses, *The 8th International Conference on Artificial Immune Systems (ICARIS)*, Volume 5666-014, 2009

Book Chapters
Using Optimisation and Machine Learning to Validate the Value of Infrastructure Investments, R. Garcia-Flores, S. Banerjee, G. Mathews (2016) Book chapter in Infrastructure Investments: Politics, Barriers and Economic Consequences

HONOURS AND AWARDS
1) 2010 Student Award for Innovation in Informatics (the award is given to a University of New Mexico graduate or undergraduate student for the best paper describing innovation or research in the field of biomedical informatics)
2) Distinction and 5th rank in Nagpur University (out of 2000 examinees) in Bachelor of Engineering program
3) Best poster award at the 1st International Online Conference of the Journal Philosophies, 2025

GRANTS
1) Grant from OpenAI Researcher Access Program for API credits (April 2024): 3000 US dollars
2) Co-investigator on two pilot grants from the AI@CAM initiative at the University of Cambridge (February 2024): 150,000 pounds
3) Grants under review: Co-investigator on MRC new investigator grant (May 2025)

INVITED TALKS
1) Responsible AI and involving patients in AI model building, Nokia Bell Labs, Cambridge, February 2025
2) Talk at UK-India Cooperation Towards a Fair AI Horizon, December 2024
3) Keynote presentation, Physics of Self-Organization in Complex Systems Satellite meeting, Conference on Complex Systems, Exeter, UK, September 2024
4) Computational Immunology, Microsoft Research, Cambridge, UK, March 2017
5) Modeling Emerging Pathogens under Uncertainty and Sparse Experimental Data, Harvard Medical School, Boston, USA, September 2014
6) Modeling Emerging Pathogens under Uncertainty and Sparse Experimental Data, IBM Research, India, August 2014
7) Computational Screens for Novel Gut Microbial Bioactive Compounds, Novartis Institutes for Biomedical Research, Boston, USA, July 2014
8) Scaling in the Immune System, Commonwealth Scientific and Industrial Research Organisation, Australia, March 2013
9) A Mathematical Model of Body Size Effects on Pathogen Replication and Immune System Response, International Network of Theoretical Immunology, Los Alamos National Laboratories, USA, August 2010

10) A Hybrid Agent Based and Differential Equation Model of Body Size Effects on Pathogen Replication and Immune System Response, School of Health Sciences, University of New Mexico, USA, April 2010

SUPERVISIONS

I have supervised 17 MPhil (postgraduate) students and 3 summer interns. I have co-supervised 1 PhD student.

ACADEMIC SERVICE

1) Program committee member of ICLR Workshop HCAIR 2026
2) Served on recruitment panels for recruiting PhD and MPhil/MSc students
3) Program committee member of International Conference on Artificial Immune Systems
4) Program committee member for ECML-PKDD Workshop on Hybrid Human-Machine Learning and Decision Making, 2023
5) Review Editor for Frontiers in Virology
6) Reviewer for IEEE Transactions on Systems, Man, and Cybernetics and IEEE Transactions on Emerging Topics in Computational Intelligence, AAAI workshop NeurMAD
7) Reviewer for SIMIODE (Systemic Initiative for Modeling Investigations and Opportunities with Differential Equations)

COMPUTER LANGUAGES

1) Programming - Python, R, MATLAB, UNIX shell scripting, C, C++, Perl, ASP.NET, Haskell
2) Databases - MS SQL Server, Sybase
3) Image analysis and automated cell tracking tools – ImageJ, CellProfiler
4) Distinctions – Ranked within the Top 250 worldwide as an open-source code contributor in the MATLAB Central code repository
https://sites.google.com/site/neelsoumya/software
5) R packages I have written
https://github.com/neelsoumya/dsSurvival

ATTENDED WORKSHOPS AND SHORT COURSES

1) Advancing Educational Practice Programme (AEPP), University of Cambridge, Sept 2023 to June 2024 (led to a Fellowship of the Higher Education Academy)
2) AI Safety Policy Fellowship, Cambridge AI Safety Hub, October 2023
3) Summer School on Data Science, Northwestern University, April 2020
4) Workshop on Spatial Statistics, Harvard University, July 2014
5) Santa Fe Institute Complex Systems Summer School, June 2nd to June 27th, 2008
6) 3rd Annual Summer School in Computational Immunology, Yale University, August 11th to August 16th, 2008
7) Gordon Research Conference in Metabolic Basis of Ecology, Maine, July 2008
8) Scaling in Biological and Social Networks Workshop, Santa Fe Institute, July 2007